HILB^D OF A CURVE IS SMOOTH PROPER OF DIMENSION D WITH DEGREE FORMULA

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 848-868 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 870-909, the complete author proof (38 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 11 of queue_pic.json (pb=1594, stmt 848-868, proof 870-909).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 0B98: sources/stacks-pic/pinned-original-source.tex, source lines 381–385. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B9H: sources/stacks-pic/pinned-original-source.tex, source lines 819–824. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B9G: sources/stacks-pic/pinned-original-source.tex, source lines 786–792. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 03GN: sources/stacks-pic/pinned-original-source.tex, source lines 909–909. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B9A: sources/stacks-pic/pinned-original-source.tex, source lines 516–522. Cited by the retained statement or proof but not

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part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B9B: sources/stacks-pic/pinned-original-source.tex, source lines 589–601. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 038F: sources/stacks-pic/pinned-original-source.tex, source lines 909–909. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0A26: sources/stacks-pic/pinned-original-source.tex, source lines 909–909. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let X be a geometrically irreducible smooth proper curve over a field k.*

(1) *The functors $\mathrm{Hilb}^d_{X/k}$ are representable by smooth proper varieties $\underline{\mathrm{Hilb}}^d_{X/k}$ of dimension d over k.*
(2) *For a field extension k'/k the k'-rational points of $\underline{\mathrm{Hilb}}^d_{X/k}$ are in 1-to-1 bijection with effective Cartier divisors of degree d on $X_{k'}$.*
(3) *For $d_1, d_2 \geq 0$ there is a morphism*
$$\underline{\mathrm{Hilb}}^{d_1}_{X/k} \times_k \underline{\mathrm{Hilb}}^{d_2}_{X/k} \longrightarrow \underline{\mathrm{Hilb}}^{d_1+d_2}_{X/k}$$
which is finite locally free of degree $\binom{d_1+d_2}{d_1}$.

Proof. The functors $\mathrm{Hilb}^d_{X/k}$ are representable by Proposition 0B9A (see also Remark 0B9B) and the fact that X is projective (Varieties, Lemma 0A26). The schemes $\underline{\mathrm{Hilb}}^d_{X/k}$ are separated over k by Lemma 0B98. The schemes $\underline{\mathrm{Hilb}}^d_{X/k}$ are smooth over k by Lemma 0B9H. Starting with $X = \underline{\mathrm{Hilb}}^1_{X/k}$, the morphisms of Lemma 0B9G, and induction we find a morphism

$$X^d = X \times_k X \times_k \ldots \times_k X \longrightarrow \underline{\mathrm{Hilb}}^d_{X/k}, \quad (x_1, \ldots, x_d) \longrightarrow x_1 + \ldots + x_d$$

which is finite locally free of degree $d!$. Since X is proper over k, so is X^d, hence $\underline{\mathrm{Hilb}}^d_{X/k}$ is proper over k by Morphisms, Lemma 03GN. Since X is geometrically irreducible over k, the product X^d is irreducible (Varieties, Lemma 038F) hence the image is irreducible (in fact geometrically irreducible). This proves (1). Part (2) follows from the definitions. Part (3) follows from the commutative diagram

$$\begin{array}{ccc} X^{d_1} \times_k X^{d_2} & = & X^{d_1+d_2} \\ \downarrow & & \downarrow \\ \underline{\mathrm{Hilb}}^{d_1}_{X/k} \times_k \underline{\mathrm{Hilb}}^{d_2}_{X/k} & \longrightarrow & \underline{\mathrm{Hilb}}^{d_1+d_2}_{X/k} \end{array}$$

and multiplicativity of degrees of finite locally free morphisms. □